

nastran95ase User's Manual

Halo One's fork of NASA's NASTRAN-95 (the April 1995 COSMIC release, NASA Open Source Agreement 1.3). This manual is for an engineer running decks. It is compiled from the fork's engineering log (HALO.md), the sources in msc/, bin/nastrn.f.in and mis/, and the VehicleDesign skills nastran-analysis, nastran-flutter, nastran95-performance and build-nastran95. Where a statement rests on one file, that file is named so it can be checked. Where the sources say nothing, the text says "not documented". The companion nastran95ase_theoretical_manual.md says what the numbers mean.

1. The executables

executable	reads	use it for
nastran95ase (nastran95ase.exe on Windows)	MSC Nastran's dialect: SOL 1xx, free, small and large field, nested and split INCLUDE, EIGRL, RBAR, RBE2, CBUSH, PBARL, CQUAD4 / CTRIA3 / PSHELL, PARAM, AUTOSPC, ASSIGN OUTPUT4, MSC launcher keywords, SOL 200	every deck written for MSC or Simcenter Nastran, including all of VehicleDesign's ASE decks
nastran95 (nastran95.exe)	the 1970s COSMIC input only: ID / APP / SOL n, 0 / TIME, fixed 8-column bulk data, no INCLUDE	NASA's 132 demonstration decks (inp/), decks written in that dialect

Both are built from one library and one main-program template, bin/nastrn.f.in, configured twice (NASTRAN_ASE_MODE in CMakeLists.txt). The front end of nastran95ase rewrites the deck before NASA's solver sees it and rewrites the print file after the solver is done; NASA's solver reads what it always read (HALO.md, "The MSC dialect front end"). Given an MSC deck, nastran95 stops on the first executive control line and the exit handler says to use nastran95ase (mds/hexit.f; the start-up pass notices a three-digit SOL or an INCLUDE). Both print the same MSC-layout print file.

Both executables are statically linked: on Windows they import only KERNEL32.dll and msvcrt.dll; on Linux they are built -no-pie so the 1 GB open core sits below 2 GB (skill build-nastran95). The rigid format library is compiled in (utility/hrfgen.f generates hrf1ib.f from rf/) and written into the scratch directory at start-up, so nothing has to be installed beside the executable (HALO.md, "Making the executable stand alone").

--version prints NASTRAN_BUILD_ID, a configure-time string naming the commit, compiler and date (HVERS in bin/nastrn.f.in). Since 2026-09-26 the Windows and Linux executables build from one fork branch, halo-ase-sol145-perf (skill build-nastran95).

2. Command line

```
nastran95ase deck.dat [output_dir] [keyword=value ...]
nastran95ase --cosmic deck.inp [output_dir]
nastran95ase --version          (also -v)
nastran95ase --help            (also -h, /?)
nastran95ase < deck.inp > deck.out    (NASA's original stdin/stdout contract, unchanged)
```

Argument handling is in bin/nastrn.f.in:

- The first argument is the deck. It is opened on unit 5; <stem>.out (the print file) and <stem>.log (the module log) go beside it, or into the output directory. Punch, plot, dictionary and checkpoint files appear only if the deck asks for them (HUSAGE).

- A bare second argument is the output directory. An argument containing `=` is a launcher keyword (next section).
- No argument on a terminal prints the usage and exits 1 instead of waiting on stdin.
- A missing deck: `nastran: deck not found: <path>`, exit 1.
- The deck stem must be 60 characters or shorter, because every output name lives in a `CHARACTER*72` and `<stem>.nptp` must fit: `nastran: deck name too long`, exit 1.
- `--cosmic` reads the deck as it is, whichever executable this is, without translation. The SOL 200 driver and the SOL 145 driver run their child analyses this way on decks they have already translated.

2.1 Launcher keywords

The scripts `VehicleDesign` carries call the solver as `nastran deck.dat scr=yes bat=no old=no append=no out=DIR`, so MSC's keywords are accepted (`bin/nastrn.f.in`, label 462):

keyword	effect
<code>out=<dir></code>	the output directory (honoured)
<code>scr=no</code> (or <code>scratch=no</code>)	keep the checkpoint: the translated deck gets <code>CHKPNT YES,DISK</code> , the problem tape <code><stem>.nptp</code> and the dictionary <code><stem>.dic</code> stay beside the print file (UIM 9464). <code>CHKPNT YES</code> without <code>DISK</code> is UFM 508 on this build (<code>HALO.md</code> , "The modes once")
<code>restart=<modes deck></code>	start a run off that deck's checkpoint: the flutter deck off its modes (section 7.6)
<code>optp=<dir></code>	where the modes run left <code><stem>.nptp</code> and <code><stem>.dic</code> , when that run had an output directory. <code>optp=<file></code> alone names the old problem tape of a <code>--cosmic</code> run, which is how the flutter children get it
anything else with <code>=</code> (<code>bat=</code> , <code>old=</code> , <code>append=</code> , ...)	<code>nastran: <arg></code> is an MSC launcher keyword; accepted and ignored on standard error

2.2 What a run leaves behind

file	what
<code><stem>.out</code>	the print file, rewritten into MSC's layout (section 11)
<code><stem>.log</code>	NASA's module log (<code>BEGN/END</code> lines per DMAP module). Its CPU column is the process's <i>system</i> time (<code>mds/cputim.f</code> returns <code>ETIME</code> 's second element, as NASA shipped it, kept because a real clock would arm the <code>TIME</code> card's limit) and its wall column is zero; do not profile from it (skill <code>nastran-flutter</code> §7, skill <code>nastran95-performance</code> §3)
<code><stem>_n95.dat</code>	the translated deck, the one actually solved (<code>bin/nastrn.f.in</code> line 298). Read it when in doubt
<code><stem>_xlat.txt</code>	the front end's messages: every card translated, dropped or renumbered, and the summary line (<code>msc_msg_open</code> in <code>msc/mscmsg.c</code>)
<code><stem>.nptp</code> , <code><stem>.dic</code>	checkpoint tape and dictionary, with <code>scr=no</code> only
<code><stem>.pch</code> , <code>.plt</code> , <code>.dic</code> , <code>.nptp</code>	removed by the exit handler when left empty (<code>mds/hexit.f</code>)
<code>s<subcase>/</code>	the SOL 145 driver's child directories, one per subcase (section 7.2)
<code><name>.op4</code> , <code><name>_s<subcase>.op4</code> , <code>.phg</code> , <code>.mgg</code>	the <code>OUTPUT4</code> files the deck asked for (section 9)

VehicleDesign gitignores *_n95.dat, *_xlat.txt, .nptp, .dic, s<subcase>/ and solver outputs; do not commit them (skill nastran-analysis §1, §8).

2.3 Scratch

Each run works in a scratch directory of its own, n95_<pid>: under %TEMP% on Windows (when TEMP is 50 characters or shorter, else <system drive>\n95tmp), under \$TMPDIR on Linux when that is set and at most 50 characters, else /tmp (mds/hoswin.f, mds/hosunx.f; HALO.md). A monarch flutter child uses 1-4 GB of it. On a machine whose /tmp is a tmpfs (RAM), runs side by side fill it and stop with GINO I/O errors; set TMPDIR to a folder on disk (skill nastran95-performance §5).

The directory is removed on every end the run lives through: a normal end and a fatal (HCLEAN, called from PEXIT and from the atexit handler), the watchdog's _exit, and on Linux SIGTERM, SIGINT, SIGHUP, SIGSEGV, SIGBUS, SIGFPE and SIGILL (msc/mscwatch.c, msc_remove_scratch, async-signal-safe getdents64/unlinkat/rmdir). The SOL 145 driver removes each child's directory after waiting for it, by the exact path the child wrote into s<id>/n95_scratch.txt (the file N95_SCRATCH_NOTE names), which covers a child that was SIGKILLED or, on Windows, ended with files open; on Windows the driver's console handler does the same on Ctrl-Break or a closed window. What still leaves a directory: a SIGKILLED or crashed top-level run, and a Windows top-level run that ends with files open. Only the run's own directory is ever removed, never a pattern and never a DIRCTY from the environment. N95_KEEP_SCRATCH=1 keeps everything (HALO.md, "The scratch directory on the ways out the exit handler misses").

3. Environment variables

All are optional. Each default applies only to a blank, so a caller that sets a variable is unaffected (bin/nastrn.f.in). The complete set of N95_ names in the tree is listed here; nothing else is read.

3.1 Run control

variable	meaning	source
N95_TIMEOUT	the wall-clock watchdog limit in minutes; 0 disables. Default: for nastran95ase 30 minutes; for nastran95 the TIME card, or NASA's 5 minutes when the card is missing. When it fires the process ends with _exit(2) and the message nastran: stopped after N minutes of wall clock, the limit for this run naming which limit applied; the print file stops where the solver was. NASTRAN's own TIME card is checked only between modules, which is why the watchdog exists	msc/mscwatch.c, bin/nastrn.f.in
N95_JOBS	how many SOL 145 children run at once; default the processor count (sched_getaffinity on Linux, the affinity mask on Windows)	msc/mscflut.c
N95_KEEP_SCRATCH	1 keeps every scratch directory	msc/mscwatch.c
N95_KEEP_OP4	1 keeps NASTRAN-95's raw OUTPUT4 file (op4_unit<n>.tmp) next to the converted one	msc/mscop4.c
N95_AERO_CACHE	a directory in which AMG keeps each (Mach, k) pair's AJJ and AMP its LU factors, content-addressed; off when unset (section 7.8)	msc/mscaec.c

variable	meaning	source
N95_DLM_QUARTIC	1 the quartic doublet-lattice kernel, 0 NASA's parabolic. The front end sets it from the deck's NASTRAN SYSTEM(270)=1 (or QUARTICDLM=1) and leaves a value the user set alone (UIM 9470); the SOL 145 children inherit it	msc/mscread.c, msc/mscisa.c
OCMEM, DBMEM	the split of open core between the modules and the in-memory database, in words. nastran95 defaults to 2,000,000 / 12,000,000 (NASA's); nastran95ase and a SOL 145 child (N95_CHILD set) give the modules three quarters; a flutter run (rigid format AERO 10, or any child of the driver) runs with no in-memory database and the whole of open core for the modules, and DBMEM turns it back on. An OCMEM above the build's open core says so on standard error and takes the default split (it used to be MESSAGE -61, END OF JOB and exit 0 with nothing solved). The build's open core is 256,000,000 words, 1 GB, set by -DNASTRAN_OPEN_CORE_WORDS (the CMake default in the file is NASA's 14,000,000; the build scripts pass 256,000,000)	bin/nastrn.f.in, skill build-nastran95
DIRCTY	use this scratch directory and leave it in place	bin/nastrn.f.in
RFDIR	a rigid format directory, overriding the compiled-in library; must fit a CHARACTER*72	bin/nastrn.f.in, mds/rfopen.f
TMPDIR (Linux) / TEMP (Windows)	where n95_<pid> is made	mds/hosunx.f, mds/hoswin.f
LOGNM, PUNCHNM, PLTNM, DICTNM, NPTPNM, OPTPNM, FTN11..FTN21, SOF1..SOF10	NASA's file-name variables, all defaulted; FTN11..FTN21 are the OUTPUT4 units the front end names itself	bin/nastrn.f.in (HUSAGE)
GFORTRAN_UNBUFFERED_ALL=y	the Fortran runtime's buffering off, so the module log's BEGN / END lines can be timed as they appear	skill nastran-flutter §7

3.2 Threads and the speed-ups

Every speed-up has a switch that turns it off; with all of them off the executable prints what Jon Chaconas's halo-ase-sol145 build prints, line for line (skill nastran95-performance §3, §4).

variable	meaning	source
OMP_NUM_THREADS	the OpenMP thread count. The SOL 145 driver sets it to every processor for each child unless the caller set it, with OMP_WAIT_POLICY=PASSIVE	msc/mscflut.c
OMP_WAIT_POLICY	PASSIVE is what the driver sets; spinning waits took 400 s against 87 s on the five-Mach deck. On Windows (MinGW libgomp) the barrier never spins and the variable does nothing	msc/mscflut.c, skill build-nastran95
N95_PK_THREADS	1 keeps the serial PK solve (one flutter loop at a time); unset or 0 takes the OpenMP default. DIAG 39 and a basis of fewer than four modes also keep it serial	mis/fa1pke.f, mis/fa1pkp.f
N95_DLM_THREADS	1 keeps GEND's serial loop over the doublet-lattice rows (and turns the batched AMG off)	mis/gendp.f
N95_AMG_BATCH	the number of (Mach, k) pairs of one Mach the doublet lattice does together; default 8, at most 16; 0 keeps AMG's own loop	mis/amgk.f
N95_AMP_PIPE	0 keeps AMP's serial pair loop	mis/ampk.f

variable	meaning	source
N95_AMP_SLOTS, N95_AMP_MB	pairs held in core by AMP's pipeline (default threads + 1, at most 16, and what N95_AMP_MB holds at about 26 NJ ² bytes a slot; N95_AMP_MB default 4096). The driver sets N95_AMP_MB to half the available memory shared among the children, between 512 MB and 2 GB, unless set	mis/ampk.f, msc/mscflut.c
N95_AJJ_SOLVE	BUILTIN (or builtin) solves the in-core AJJ with the fork's own LU instead of LAPACK's ZGETRF/ZGETRS. The LAPACK solve is the one change that is not bit-identical: a blocked LU rounds differently and marginal PK roots move (108 of 170,034 print lines on the monarch; no crossing)	mis/ampczs/, skill nastran95-performance §4
N95_BLAS_THREADS	OpenBLAS's thread count, set explicitly because a static OpenBLAS otherwise runs on one thread; the driver sets it to processors / children	mis/ampczs/ampczt_openblas.f, msc/mscflut.c
N95_INCORE_AJJ	0 turns off the in-core AJJ solve (AMPC's and AMPF's) and runs NASA's out-of-core TRANP1 / CFACTR / CFBSOR path, hours per Mach on the monarch; the way to A/B the two paths	mis/ampcz.f
N95_ISA	0 keeps the baseline x86-64 build of the PK QR, EGNVCT and the kernel; otherwise their -march=x86-64-v3 copies run when the processor has AVX2 (same bits, more speed)	msc/mscisa.c
N95_PK_SECANT	1 replaces NASA's fixed-point step in the PK iteration with a secant step; off by default because it moves bits (section 7.5)	mis/fa1pkp.f, mis/fa1pke.f
N95_PK_STATS	1 prints, per child, a histogram of QR solves per root to standard error at the end of the PK loop; moves no bits	mis/fa1pkp.f
N95_THROTTLE (Windows)	1 keeps Windows' execution-speed power throttling; by default the executable opts out (SetProcessInformation)	msc/mscwatch.c, skill build-nastran95
N95_CHILD, N95_SCRATCH_NOTE	set by the SOL 145 driver for its children (the ASE open-core split; the scratch-path note); not for users	msc/mscflut.c, bin/nastrn.f.in

N95_NO_ISA_V3 is a compile-time define, not an environment variable (msc/mscisa.c).

4. Solution sequences

The SOL map is `sols[]` in `msc/mscexec.c`:

MSC SOL	COSMIC APP and rigid format	analysis
101 (SESTATIC)	DISPLACEMENT 1	linear statics
103 (SEMODES)	DISPLACEMENT 3	normal modes
105 (SEBUCKL)	DISPLACEMENT 5	buckling
107 .. 112	DISPLACEMENT 7 .. 12	direct/modal complex eigenvalues, frequency and transient response

MSC SOL	COSMIC APP and rigid format	analysis
145 (SEFLUTTR)	AERO 10	modal flutter (section 7)
146 (SEAERO)	AERO 11	modal aeroelastic (gust) response (section 8)
200	not translated: a design cycle run by <code>msc/mscopt.c</code> around child analyses of this executable	design optimisation (section 4.1)
1, 3	DISPLACEMENT 1, 3	the bare rigid format numbers

SOL 106, 129, 144, 153, 159, 400, 401, 402 are named in `no_map[]` so the fatal (UFM 9110) can say what was asked for: "NASTRAN-95 has no equivalent". There is no COSMIC counterpart to SOL 144, static aeroelastic trim; the NASTRAN trim decks need the licensed solver (skill `nastran-flutter` §7). A deck naming no solution the front end recognises is UFM 9111.

4.1 SOL 200

Implemented in `msc/mscopt.c` and `msc/mscopt2.c`: `DESVAR`, `DVPREL1`, `DVMREL1`, `DLINK`, `DRESP1` (WEIGHT, VOLUME, `FREQ`, `EIGN`, `DISP`, `STRESS`), `DCONSTR`, `DCONADD`, `DSCREEN`, `DOPTPRM`, `DESOBJ`/`DESSUB`/`DEGLB`, `ANALYSIS = STATICS|MODES|BUCK`. Sensitivities are forward differences from child `--cosmic` runs, one per free design variable per cycle; the approximation is convex linearisation (CONLIN) solved through its dual; weight and volume are closed-form from the model. Anything else in the design model is a numbered fatal naming the card (no `DRESP2`, `DEQATN`, `DVGRID`). The report is in `<stem>.out`, the final design in `<stem>_final.dat`. On MSC's three-bar truss example it follows MSC's design-cycle history to within half a percent at every cycle (`HALO.md`; skill `nastran-analysis` §6).

5. What the translator does to a deck

The front end (`msc/mscread.c` reads, `msc/mscexec.c` translates executive and case control, `msc/mscxlat.c` the bulk data, `msc/mscwrite.c` writes the 8-column deck) records every action in `<stem>_xlat.txt` and on the terminal, and ends with UIM 9001: SOL n became rigid format r (APP ...). c cards written, d dropped, r ids renumbered, k degrees of freedom constrained for want of stiffness.

5.1 Reading

Small field (8 columns, continuation marker in 73-80), large field (name ending in *, 16-column fields, four per line) and free field (commas) are read, and a deck may mix them. A line contributes a fixed number of field slots - eight in small and free field, four in large field - whether or not they were written. Nested `INCLUDEs` (UFM 9011 past the nesting limit) and the `INCLUDE 'path / rest'` form split over two lines are read. A card's fields grow as needed (the cap was 256 fields until fork `d5e0e5f`, which silently truncated the 501-point `TABLED1` of the turbulence decks; it is now 1,048,576).

5.2 Executive control

SOL becomes APP + SOL r,0 + TIME. NASTRAN `SYSTEM(270)=1` or `QUARTICDLM=1` sets `N95_DLM_QUARTIC` (UIM 9470 says which kernel runs). `ASSIGN OUTPUT4` is read (section 9); other `ASSIGN` statements are ignored with UIM 9121. `NASTRAN BANDIT=-1` is written above 300 grids (UIM 9113), because the bandwidth resequencer's scratch array stops a run with SFM 3037 on a few thousand grids with rigid elements. `MAXLINES = 999999` is added.

5.3 Case control

Commands NASTRAN-95 reads with the same meaning are kept (`case_keep[]` in `msc/mscexec.c`): `TITLE`, `SUBTITLE`, `LABEL`, `ECHO`, `MAXLINES`, `LINES`, `SPC`, `MPC`, `LOAD`, `DEFORM`, `TEMPERATURE`, `METHOD`, `CMETHOD`, `FMETHOD`, `SDAMPING`, `FREQUENCY`, `OFREQUENCY`, `TSTEP`, `DLOAD`, `IC`, `NONLINEAR`, `GUST`, `RANDOM`, the `XY*` plot commands, `DISPLACEMENT`, `VELOCITY`, `ACCELERATION`, `SPCFORCES`, `OLOAD`, `AEROF`, `STRESS`, `ELFORCE`, `FORCE`, `SET`, `SUBCASE`, `SUBCOM`, `SUBSEQ`, `SYMMETRY`, `REPCASE`, `OUTPUT`, `AXISYMMETRIC`, `MODES`, `SVECTOR`, `THERMAL`, `FLUX`, `TRIM`, and since 2026-09-26 `K2PP`, `B2PP`, `M2PP`, `TFL` (direct matrix input; before that they were dropped and a `DMIG` never reached the solver). Names are prefix-matched. Output options in parentheses are MSC's and are dropped while the request is kept (UIM 9103).

Commands that are MSC's are dropped, each with what its absence costs named (UIM 9101): `RESVEC` (MSC adds residual vectors by default, NASTRAN-95 never did), `ANALYSIS`, `WEIGHTCHECK`, `GROUNDCHECK`, `AUTOSPC` (the front end applies the equivalent constraints), `MEFFMASS`, `MODALSE`, `SDISPLACEMENT` (UFM 614 in the solver), `ESE`, `ECHOON`, `ECHOOFF`, `DESOBJ`, `DESSUB`, `DESGLB`, `DSAPRT`, `SEALL`, `SUPER`, `K2GG`, `M2GG`, `B2GG`. A command the front end does not know is left out with UWM 9102. `APRES` becomes `AEROF` (NASTRAN-95 prints pressures and forces together), or is dropped when `AEROF` is already asked for (UIM 9104). `SET n = ...` is written as a command with its continuation lines. Ids in `SET` lists and `XY` requests follow the bulk data's renumbering (section 5.5); a `THRU` range bounded by a renumbered id cannot be followed (UWM 9105).

A SOL 103 deck whose case control asks for no output gets `DISPLACEMENT = ALL` (UIM 9114).

5.4 Bulk data, card by card

Dispatch is the loop at `msc/mscxlat.c` line 1025.

MSC card	becomes	notes
<code>RBAR</code> , <code>RBE2</code>	<code>CRIGD1</code>	the independent end is chosen so that a grid on a <code>SUPPORT</code> or <code>SPC</code> is never made dependent; the element id is kept (so UFM 2192 names the <code>RBAR</code> 's id)
<code>CBUSH</code> + <code>PBUSH</code>	six <code>CELAS2</code> when the two grids are coincident; <code>CONROD</code> carrying <code>K1</code> alone when they are separated	a separated <code>CBUSH</code> with <code>K2..K6</code> set has no equivalent and is UFM 9206, not a guess
<code>EIGRL</code>	<code>EIGR FEER</code> for ND roots about <code>V1</code> Hz, or 0.5 Hz when <code>V1</code> is blank (UIM 9211)	a zero shift on a free-free model loses every mode; never write <code>V1 = 0.0</code>
<code>EIGC CLAN</code> (or any MSC method)	<code>EIGC HESS</code> (UIM 9116)	the continuation's ND says how many roots to keep
<code>PBARL</code>	<code>PBAR</code> with <code>A</code> , <code>I1</code> , <code>I2</code> , <code>J</code> worked out from the dimensions (UIM 9209)	

MSC card	becomes	notes
CBEAM, PBEAM, PBEAML	not handled	the header comment of <code>msc/mscxlat.c</code> and the catalogue's entry 307 say they become CBAR / PBAR at the end-A properties (and <code>msc/mscmsg.c</code> 's example message is a UFM 9203 for a non-prismatic CBEAM), but the dispatch loop has no handler for them and they are not in <code>copy_cards[]</code> , so they fall to the "anything else" row below; the <code>nastran-analysis</code> skill reports CBEAM as UFM 9299. Write CBAR / PBAR
CQUAD4, CQUAD, CQUADR; CTRIA3, CTRIAR; PSHELL	CQUAD2, CTRIA2; PQUAD2 / PTRIA2	panels thinner than 1e-6 are dropped as massless drawing aids ("dropped: no stiffness")
SUPPORT1	SUPPORT (UIM 9214)	NASTRAN-95 has no set-selected form; the reference freedoms are always in force
SPC1	SPC1 with THRU expanded and the grid list made contiguous	
FLUTTER	PKNL becomes PK + PARAM PKMATCH 1 (UIM 9123); IMETH other than L becomes L (UIM 9117)	TCUB is UFM 316 in Simcenter too; the repo's decks write L
SPLINE1	SPLINE1 without METH/USAGE	
TABDMP1	TABDMP1 without TYPE (G only)	
PARAM	WTMASS, COUPMASS, GRDPNT, G, W3, W4, LMODES, LFREQ, HFREQ, MAXRATIO, Q, MACH, KDAMP, IFTM pass through; PKMATCH, PKVECT pass through (this fork's); GUSTAERO's sign is flipped (UIM 9119: +1 computes the gust aerodynamics in NASTRAN-95, -1 in MSC); every other PARAM (POST, AUTOSPC, K6ROT, RESVEC, ...) is "dropped: MSC only"	KDAMP passes unchanged but means the opposite in the two codes (section 8)
CONM2	unchanged; the ones with no rotary inertia (fields 9-14 blank or zero) are counted and reported on a modal solution (UWM 9133)	
passed through unchanged (<code>copy_cards[]</code> : "cards NASTRAN-95 reads with the same name, fields and meaning")	GRID, GRDSET, SPOINT, EPOINT, SEQGP, CORD1R/C/S, CORD2R/C/S, MAT1-MAT5, PBAR, PROD, PTUBE, PSHEAR, PELAS, PDAMP, PMASS, PVISC, CBAR, CROD, CTUBE, CONROD, CSHEAR, CVISC, CELAS1-4, CDAMP1-4, CMASS1-4, CONM1, CONM2, MPC, MPCADD, SPC, SPCADD, SPCD, SUPPORT, OMIT, OMIT1, ASET, ASET1, the static loads (FORCE*, MOMENT*, GRAV, LOAD, PLOAD*), temperatures, EIGR, EIGB, EIGP, FREQ*, TSTEP, TLOAD1/2, RLOAD1/2, DLOAD, DAREA, DELAY, DPHASE, TABLED1-4, TABLEM1-4, TABLES1, TABLEST, TF, TIC, NOLIN1-4, AERO, AEROS, CAERO1, PAERO1, SPLINE2, SET1, FLFACT, MKAERO1, MKAERO2, TRIM, AESTAT, AESURF, GUST, RANDPS, TABRNDG, TABRND1, DMI, DMIG	

MSC card	becomes	notes
dropped as drawing or post-processing aids	PLOTEL , PBUSHT , RESVEC , MDLPRM , CBARAO , MONPNT1 (drop_cards [])	
anything else	UFM 9299 naming the card, file and line; the deck is not translated, nothing is solved, no print file	CBEAM lands here (above); CAERO4 is not in the front end (skill solver-benchmarks)

A SOL 145 or 146 deck with several subcases in a single run (not through the driver) gets UWM 9115: NASTRAN-95's rigid format solves the first and the rest are left out.

5.5 Renumbering and the automatic constraints

NASTRAN-95 packs an id and a component into one 32-bit word, so every id above $2^{24} - 1 = 16,777,215$ is renumbered, counting down, everywhere it is referenced (renumber_ids in msc/mscxlat.c; UIM 9302 gives the count). PARAM,GRDPNT moves with it (missing that put the two codes' centres of gravity 0.15 m apart on a 1,400-grid model with mass and inertia agreeing), as do DAREA / DELAY / DPHASE / TIC grid ids, SET lists and XY points. The print file puts the deck's numbers back (msc/mscf06.c). The aero reference grid 99999999 of VehicleDesign's decks becomes 16777215 in <stem>_n95.dat.

Every degree of freedom nothing is attached to is constrained in an SPC1 set (9998 in the repo's runs) joined to the user's SPC selection through an SPCADD, which is what MSC's AUTOSPC does (UIM 9301). NASTRAN-95's own PARAM AUTOSPC 1 also exists and prints UIM 2435 with the cards it generated; what NASTRAN-95 lacks is MSC's PARAM,AUTOSPC,YES spelling (HALO.md, the catalogue audit).

5.6 Writing

Small field, eight columns, continuations tagged +0000001 on (a restart run's begin with a letter, +R000001). A real is written in the eight-column spelling that reads back closest to the value, NASTRAN's exponent form with the E dropped (1.23457-5) included; a number that does not fit is re-spelled, never cut (-6.89e+04 cut to eight characters reads as -6.89: this happened on a PBAR). Large-field cards are used when even that loses more than 1e-5 (msc_r8 in msc/mscwrite.c).

6. Normal modes (SOL 103)

The eigensolver is FEER, with a shift from the EIGRL. Points to know when running the repo's modal decks (skill nastran-analysis §2-§5):

- Budget: mfp / monarch_gvt decks 20 s at 120 modes; the CAD-mass configs (4,500-5,400 grids) 85-95 s at 120 modes, 15-25 s at 20.
- FEER returns more roots than asked (a reduced problem of twice the size); the print rewrite and the .phg writer cut to the number requested (UIM 9131).
- On a semi-definite mass matrix - lumped CONM2s with no rotary inertia, every model in the repo - FEER's reduction can lose the mass norm of a trial vector; the fork reseeds (UWM 2394 once) and may return fewer modes than asked (UWM 2390). The modes it does return are accurate. The exit handler repeats both on the terminal; UWM 9133 from the translator warns up front with the count of such CONM2s and the cure: rotary inertia on the lumped masses, or fewer modes. The deck that showed all this (830 grids) gets 89 of 120 modes in 7 s; raising the orthogonalisation cap from 14 to 60 passes bought four rows and nothing else (HALO.md, "Never hang"). DIAG 16 prints the reduction row by row.
- A rigid format that needs a mass matrix and has none stops in its own DMAP check with no numbered message, END OF JOB and exit 0; the exit handler says so on the terminal.

- `ECHO = SORT` is what the grid reader needs; `DISP = ALL` the eigenvector reader; `PARAM,GRDPNT,<grid>` the mass reader.
- `ASSIGN OUTPUT4` with the `SEMODES` alter writes `.phg` (PHIG) and `.mzg` (MGG) in MSC's formatted layout (section 9); rigid format 3 only.

7. Flutter (SOL 145)

7.1 What the deck must say

NASA's AERO 10 rigid format has three methods, K, KE and PK, and its PK loops over every combination of the density, Mach and velocity `FLFACT` lists with a hard stop at 100 loops (Flutter Error 3). VehicleDesign's decks are matched-point analyses: at a fixed Mach, one point per altitude, density and true airspeed belonging together, which MSC and Simcenter spell `FLUTTER ... PKNL` (HALO.md, "PKNL, and PK on matched points").

- The IFP accepts `PKNL` as a fifth method name (`mis/ifs5p.f`), and the front end writes MSC's `PKNL` as `PK` with `PARAM PKMATCH 1` emitted once (`msc/mscxlat.c`); a deck that already says `PK + PKMATCH` passes through. `rf/AERO10` hands `PKMATCH` to FA1 as its fourth parameter and `PKVECT` as its fifth (`FA1 .../V,Y,PKMATCH=0/V,Y,PKVECT=0`).
- With it, FA1 builds one flutter loop per entry of the density and velocity lists taken together, the Mach list walked alongside (its last entry repeated when it is shorter) (`mis/fa1.f`, label 98). The PK iteration itself is NASA's.
- The `FLUTTER` card, in its own order: `SID`, `METHOD`, `DENS`, `MACH`, `RFREQ` (the velocity `FLFACT` for PK), `IMETH`, `NVALUE` (the roots found per loop, FA1's `NEIW`), `EPS` (the convergence tolerance on k; 0.001 when blank or zero, `mis/fa1pke.f`). The driver reads the Mach and velocity `FLFACT` ids off it (`subcase_mach`, `child_loads_velocity` in `msc/mscflut.c`).
- `PARAM LMODES` chooses the modes that carry the flutter solution; the `EIGRL` count only sets how many modes `READ` finds (the References distillation of NASA's User's Manual, §3).
- A negative velocity in the velocity `FLFACT` marks that loop for eigenvector output.
- `FMETHOD` selects the `FLUTTER` card; `METHOD` the `EIGR`; `SDAMPING` the `TABDMP1`; `CMETHOD` the `EIGC` (K method only). `OFREQUENCY` must precede `OUTPUT(XYLOT)`.

7.2 Several subcases: the driver

NASTRAN-95's flutter rigid format solves one subcase: FA1 reads the first `CASECC` record for its `FMETHOD` and never comes back for another. The repo's decks carry one subcase per Mach. `msc/mscflut.c` therefore:

- writes one translated deck per subcase and runs them as child processes of this executable (`--cosmic`, with `N95_CHILD` set for the open-core split), as many at a time as there are processors (`N95_JOBS` overrides), each in `s<subcase>/` under the output directory (UIM 9450). Two children in one directory would write over each other's fixed-name files and die in GP4, hence the directories;
- gives each child its subcase's Mach, read off the `FMETHOD`'s `FLUTTER` card and its Mach `FLFACT` (one value on the matched-point decks): `PARAM MACH` for the ADR load recovery, and the `MKAERO1 / MKAERO2` lists cut to that Mach (UIM 9455), so a five-Mach deck's children compute one Mach's aerodynamics each instead of five;
- puts the marked loop's velocity on each child's `AERO` card (UIM 9456) so the `AEROF` loads come out at that loop's reduced frequencies (section 7.4);
- sets the threads: `OMP_NUM_THREADS` = every processor for each child (the coarse work - doublet-lattice rows and PK loops - shared by the scheduler), `OMP_WAIT_POLICY=PASSIVE`, `N95_BLAS_THREADS` = processors / children for the LU, `N95_AMP_MB` as in section 3.2; a caller's values stand. Measured on the

five-Mach monarch deck on 32 processors: 102 s with an even split, 87 s this way, 400 s with spinning waits;

- waits for whichever child finishes (`waitpid` on POSIX, `WaitForMultipleObjects` on Windows), prints `nastran: subcase N -> sN/<stem>_sN.out (exit code n)` for each, removes each child's scratch, and joins the children's print files into `<stem>.out` in subcase order (`<stem>_s<subcase>.out` each), then rewrites the joined file into MSC's layout (`msc_f06`), as a single run's print is. The final line is `nastran: <stem> -> <stem>.out (N subcases joined);`
- converts every child's OUTPUT4 files into the output directory per subcase (section 9);
- exits with the worst child's code; a child that could not be started or translated (UFM 9451, 9452), ended by a signal, or crashed on Windows (a negative NTSTATUS) counts as 3. A child that left no print file is UWM 9453.

On Linux a child asks for SIGTERM when the driver dies (`PR_SET_PDEATHSIG`); on Windows every child is in a job object with kill-on-close, so a driver stopped by its watchdog or the user takes its children with it (skill `build-nastran95`). A per-point split (one child per matched point) is deliberately not offered: every child recomputes the modes and the whole aerodynamic matrix set, and the PK iteration itself takes seconds.

Before starting a run, look for a running `nastran95ase` (`pgrep -a nastran95ase`; `Get-CimInstance Win32_Process` on Windows): two runs of one deck in one directory write the same child files (skill `nastran-flutter` §7).

7.3 Marked loops, eigenvectors and PKVECT

A negative velocity in the `FLFACT` list is the request both MSC and NASTRAN-95 understand. For every accepted root of that loop FA1 prints the modal vector - `EIGENVECTOR FROM THE PK METHOD`, then `EIGENVALUE = re im LOOP = n VELOCITY = v` (the header repeated at the top of every page a vector runs over), then the complex coefficient of each mode of the basis (`mis/fa1pkv.f`, formats 21 and 26) - and the rigid format recovers the physical vectors at the `DISP` set, when the case control asks for them (`DISP` and `OFREQUENCY`; `OFREQUENCY` is here a list of velocities, matched by `MODACC` on the imaginary part), through `MODACC`, `DDR1`, `SDR2` and `OFP` as `COMPLEX EIGENVECTOR` tables in the layout MSC prints. Every root of the loop is printed, not the one that crossed: the monarch deck with twenty loops marked prints 1,201 physical vectors at 251 points, 120 MB (`HALO.md`, "Eigenvectors at marked loops").

A `DISP` set that names the `CAERO1` box ids gets one row per box in the `COMPLEX EIGENVECTOR` tables - the plunge in `T3` and the pitch in `R2`, in the box's frame - with no parameter (MSC needs `PARAM OPPHIPA` for the same rows).

`PARAM PKVECT 1` has `FA1PKV` print the modal vector of every root at every loop, while the recovery scratch (301) is written for the marked loops alone. The print grows by one complex number per mode per root per loop (the monarch at 110 modes, 40 loops and five Machs: some 22,000 vectors, about 150 MB, against 813 MB for the physical recovery of five marked loops). The flutter summaries are unchanged by it to every printed digit, and the physical vector of any point is Φq with the modes run's Φ (VehicleDesign's `animate_nastran_flutter_mode`; the reader `read_nastran_pk_eigenvectors` joins a vector over its page breaks).

7.4 Aerodynamic loads (AEROF)

The case control request is NASA's `AEROF` (MSC's `APRES` is written as `AEROF`). ADR builds the box loads of every recovered root and `ADRprt` prints them as `AERODYNAMIC LOADS (UNIT DYNAMIC PRESSURE)`, one line per box with the real and imaginary parts of $T1..T3$ and a second with $R1..R3$. ADR takes each root's reduced frequency from $BOV = b/V$, which APD forms from the `AERO` card's velocity field; the matched-point decks leave it blank, so $BOV = 0$, UIM 2272, and nothing printed - which is why the driver writes the

marked loop's velocity onto each child's AERO card. With one marked loop per subcase every recovered root's loads come out at its own k; with several, the first loop's velocity serves and UWM 9456 says the other loops' loads are at the wrong k. Until fork a262158 these tables printed about 1e-19 on every box (NASA's UNPACK limit, section 13); they now print the loads (HALO.md, "Tall columns").

7.5 Roots: NVALUE, tracking, the secant

- FA1 solves NVALUE roots per loop, numbered by acceptance in ascending frequency, not sorted by mode. On monarch_demo_asm1083 83 of the 110 modes give the same crossings as all 110 with fewer PK warnings and a fifth less time; the rule used there is NVALUE = the highest root that reaches 30 Hz anywhere in the baseline all-roots run (skill nastran95-performance §4, "The roots found").
- Root tracking (mis/fa1pkt.f, FA1PKT/FA1PKU): the roots of a loop are re-ordered so that POINT n of the summary follows POINT n of the loop before, by the modal assurance criterion between the PK eigenvectors plus a quarter of the relative eigenvalue distance, assigned greedily; the print says ROOT TRACKING: FLUTTER LOOP n - m OF k ROOTS TOOK ANOTHER PLACE TO FOLLOW THE LOOP BEFORE for every loop that reordered. The tracking restarts when the Mach changes and is skipped with a note when the work space is too small. VehicleDesign's plotter re-sorts after the fact as a check; the two agree.
- N95_PK_SECANT=1: a secant step on the residual $h(k) = k_{\text{from_root}}(k) - k$ once two iterates exist, falling back to NASA's step when it would be non-positive or more than twice NASA's. It cuts the QR solves per root (2.24 to 1.68 on the monarch at Mach 0.10, EPS 1e-3), moves the lowest crossing per Mach by at most 0.001 m/s, and lands about a tenth of the heavily damped roots on another branch, as a change of EPS does. An opt-in for the lowest crossing, not a default (HALO.md, "A secant step").
- A failure to converge in 10 iterations prints PK METHOD FIALED TO CONVERGE [spelled so in mis/fa1pke.f] with LEAST SQUARES FIT APPROXIMATION IMPLEMENTED and takes the fixed point of a line fitted through the trials.

7.6 The modes once: checkpoint and restart

Every child of the driver solves the eigenproblem again (a child is a complete run of the AERO rigid format). The alternative is NASTRAN's own checkpoint/restart (HALO.md, "The modes once"; skill nastran-flutter §7):

```
nastran95ase <config>_modes.dat scr=no                (the SOL 103 deck gen_nastran_cards writes
beside the flutter deck)
nastran95ase <config>_flutter.dat restart=<config>_modes.dat
nastran95ase <config>_modes.dat scr=no ci_out
nastran95ase <config>_flutter.dat restart=<config>_modes.dat optp=ci_out ci_out
```

scr=no checkpoints: CHPNT YES,DISK, the dictionary punched to <stem>.dic, the problem tape <stem>.nptp (about 100 MB on the monarch; mds/gnfiat.f had to be changed to honour NPTPNM, whose IPERM bit was never set). restart= reads the dictionary into the executive control after the TIME card (UIM 9462), translates the modes deck a second time and drops from the restart deck every card whose translated text that run had (a hash set), so only the cards the modes run did not have are written (UIM 9463: on the monarch, the 64 aero and flutter cards); NASTRAN merges them with the old bulk data off the tape. The tape is hard-linked (copied when the volume refuses) into the output directory as optp.nptp, because DSNAMES are CHARACTER*80 and the children run one directory down. The rigid-format-switch restart (RF3 -> AERO10, UIM 4145) skips GP1 through READ and runs GKAM / APD / GI / AMG / AMP / FA1 / FA2. The joined print gets the modes run's REAL EIGENVALUES pages spliced in after each sorted echo (msc/mscf06.c), so read_khh still reads them; UWM 9466 when the modes print is missing.

What makes a restart solve the modes anyway: any card of the restart deck that the modes run did not have and that is not an aero or flutter card - NASTRAN's restart tables take it as a change to the structure

and re-execute from GP1 (UWM 9467 names such cards). The first monarch try restarted off the vibe deck and re-solved everything (72 CELAS2 from the CBUSH springs, numbered above the highest element id, which the CAERO1 boxes move, and the 6 aero CORD2R), which is why the modes deck carries aero_model.bdf and no flutter cards. A card that replaces one of the modes run's under the same id is UFM 311 to the merge. Regenerate the modes deck whenever the structure or the aero model changes.

Flutter runs - AERO 10 out of the translation, or any child - run with no in-memory database, because NASA's DBM loses the block count of the eigenvalue file FA1 appends once the database is full (BLOCK NUMBERS INCONSISTANT ON OPEN IN DBMMGR at loop 23 of every restarted 120-mode monarch child); DBMEM turns it back on.

The restarted summaries agree with a direct run's to six significant figures on the monarch (FEER's basis off the tape against the one READ recomputes), digit for digit on the two-subcase test deck. Those last digits move which matched points' PK solutions glitch - a single point reading $g + 0.010$ between two at -0.016 and -0.017 - so the plotters require two points above the threshold (drop_isolated). On Linux the restart saves little (73.5 s direct against 72 s), since the direct run's children solve their modes side by side.

7.7 Timings

deck, machine	time	source
five-Mach monarch_demo_asm1083_flutter.dat (120 modes, 60-61 matched points, 2,282 boxes, 24-48 k per Mach), 16-core/32-thread Linux, unmodified source	19 min 50 s	HALO.md, "Linux"
the same, every speed-up, parabolic kernel	40.5 s (quartic 66-68 s)	skill nastran95-performance §1
the same, N95_AJJ_SOLVE=builtin	179-200 s, Jon's print line for line	same
the same deck on Jon's Windows build (fcd681b , 16 cores)	25.6 min	HALO.md
the committed 110-mode PKVECT deck, Core Ultra 7 165U laptop (14 threads): Jon's exe / parity build parabolic / quartic	1768.6 s / 309.9 s / 387.7 s	skill nastran95-performance §1
the reduced one-Mach deck on MSC Nastran (before the switch to Simcenter)	6.4 min	skill nastran-flutter §7
Simcenter 2606, the five-Mach deck	hours against 6.5 min for nastran95ase	VehicleDesign repo guide

Linux timings from 2026-09-26 were taken with other jobs running and are provisional. Time on a quiet machine: three runs, the median.

7.8 The aerodynamic cache

With N95_AERO_CACHE=<directory>, AMG keeps each (Mach, k) pair's AJJ under a 64-bit hash of every input it depends on (the group's ACPT record, its counts, NJ, the symmetry flags, the reference chord, Mach, k, and a kernel version word IAECV: 1 parabolic, 3 quartic) and AMP keeps each pair's LU factors and pivots under a hash of the AJJ (the LAPACK solve only; with N95_AJJ_SOLVE=builtin nothing is kept). Entries are files <key>.1 and <key>.2, written as a temporary and renamed into place so runs side by side can share the directory; nothing is ever deleted (empty the directory when the aerodynamic model is gone). The print is not told; the counts go to standard error at exit (N95_AERO_CACHE: AMG (AJJ) 48 of 48 pairs from the cache). A cached run prints the uncached print to the bit, by construction (msc/mscaec.c; HALO.md, "The aerodynamic cache"). Size: 42 MB of AJJ plus 83 MB of LU per pair at 2,282 boxes;

HALO.md gives 21 GB for the five Machs' 178 pairs, the mscaec.c header "about 19 GB". On the monarch study deck: no cache 28.3 s wall / 578 CPU s; every pair from the cache 21.5 s / 353 s.

8. Gust response (SOL 146)

AERO 11 goes through the front end since forks c5a910b-03afoaf, and every repo gust deck (von Karman PSD, 1-cos, seeded turbulence) runs to the end and matches Simcenter 2606 since fork f28bo83 (skill nastran-flutter §7). What the translator does for it: GUST, RANDPS, TABRNDG, TABRND1 copied; RANDOM, OUTPUT(XYPLT) and the XY requests written as they are, their ids renumbered with the bulk data; PARAM Q and PARAM MACH passed (one dynamic pressure and one Mach per execution); GUSTAERO's sign flipped (+1 computes the gust aerodynamics in NASTRAN-95, -1 in MSC); SDISPLACEMENT dropped; the DAREA/DELAY/DPHASE/TIC grid ids renumbered with the grids (the gust DAREA sits on the aero reference grid). PARAM KDAMP passes unchanged but its sign means the opposite in the two codes: +1 is viscous modal damping in MSC and Simcenter and complex stiffness in NASTRAN-95; both default to viscous and no repo deck sets it. Random response is available only through the XY tabular print (XYPRINT ... PSDF), per NASA's manual.

Four faults were fixed on the way (HALO.md, "Tall columns" and "SOL 146 random response"): NASA's UNPACK stopping at row 65,536 (the gust loads came out zero; doubling WG changed nothing - that is the test for it); SDR2B reading an absent SIL's stale trailer (SFM 3001 with ELFORCE); case control keeping renumbered ids (RAND2 SFM 3002); and the spline in k solved in single precision (LSPLND now solves it in double; the PSD response was jagged above 1.5 Hz before). The monarch gust decks run in about a minute on Linux.

9. OUTPUT4

msc/mscop4.c honours ASSIGN OUTPUT4='<file>' ... UNIT=n FORM=FORMATTED plus an OUTPUT4 alter, for rigid formats 3, 10 and 11 only (UWM 9125 otherwise):

- MSC alters by label; NASTRAN-95 by DMAP statement number. In rigid format 3 the alter is written after statement 77 (SDR1, where PHIG and MGG both exist; the number from a DIAG 14 listing of the 1995 executable). In AERO 10 and AERO 11 it is written after statement 90 (AMP), where QHHL, QHJL, QKHL, MHH, KHH, BHH and PHIDH all exist. The alter text in the deck is otherwise ignored: a Simcenter deck's ALTER 'A3J0,WSKJF' / OUTPUT4 QHH,, ,//0/101/2 maps QHH, QHJ, QKH to the whole lists QHHL, QHJL, QKHL, so one block serves both solvers.
- Data blocks known (dbmap[]): PHG/PHIG, MGG, KGG, PHA/PHIA, MAA, KAA, QHHL, QHJL, QKHL, QHH, QHJ, QKH, MHH, KHH, BHH, PHDH/PHIDH. Any other is UFM 9122.
- MSC's units (101, 102, ...) are mapped to Fortran units 11, 12, ... in order of appearance; up to 8 requests (OP4_MAX). The main program opens units 11, 12 and 13-21 (FTN11..FTN21); mscop4.c's header speaks of units 11 to 24 (FTN11..FTN24), which the main program does not open. NASTRAN-95 writes op4_unit<n>.tmp, rewritten on the way out into MSC's formatted layout: header NCOL NROW FORM TYPE (4I8), name (A8), 1P, 5E16.9; real matrices out as type 2 and complex as type 4 with real, imaginary pairs; per column IC IR NW (3I8) then the values five to a line; null columns left out (as MSC's own OUTPUT4 does); a trailer column NCOL+1 1 1 and one 1.0; a negative (symmetric) NASTRAN-95 form made 6. PHIG/PHIA are cut to the mode count the print file was (UIM 9131). N95_KEEP_OP4=1 keeps the raw file.
- The SOL 145 driver converts each child's files into the output directory as <name>_s<subcase>.<ext> (UIM 9133), so each Mach's QHHL is its own file, n_h rows by n_h x n_k columns, one block per k; MHH/KHH/BHH come out per subcase too though they are the same in every child. UWM 9134 when no child wrote the file.

- **The order of the blocks is the order of the (Mach, k) list**, and NASA's APD wrote that list card by card in sorted bulk data order, so a k list split over several MKAERO1 cards came out shuffled (a two-card deck put k = 1.6, 3.2 ahead of 0.01 .. 0.8; Simcenter does the same and does not sort). The flutter and gust interpolations never cared; a reader of the exported blocks that assumed ascending k built a plant whose roots were nothing like PK. APD now sorts the pairs by Mach then k (APDMKS in mis/apd.f) and prints the list it used as UIM 9457, per child; VehicleDesign's read_nastran_qhh1 takes the order off that print.
- **Known (2026-10-01): the export needs a cold flutter run.** A flutter deck restarted off the modes checkpoint is a modified restart with a rigid-format switch, and NASTRAN executes only the DMAP instructions its restart tables flag; an ALTERed OUTPUT4 has no table entry, so the children leave empty op4_unit*.tmp files and the driver reports "no child wrote" (UWM 9134). Each child then has to solve the modes again (85-95 s on the monarch, in parallel), which is what the export costs until the restart flags the alter.

The repo's OUTPUT4_rd.m reads the files; test_ase_plant replays FA1's arithmetic on the export and gets the printed PK roots back to 1e-4 (skill nastran-flutter §8).

10. Console verdicts and exit codes

The exit handler (mds/hexit.f, registered with atexit) reads the print file back and sets the code; the fatal explainer (msc/mscdiag.c) runs in both executables.

exit	console line	meaning
0	nastran: <stem> -> <dir><stem>.out	END OF JOB reached, no fatal
0	nastran: fewer modes than requested were found (UWM 2390 ...)	FEER reseeded and stopped early; the modes printed are accurate; the count is in the print
0	nastran: the mass matrix is singular or indefinite ... (UWM 2394)	lumped CONM2s with no rotary inertia (UWM 9133 said so up front)
0	nastran: this solution needs a mass matrix and the model has none ...	the rigid format stopped in its own check: no eigenvalue table, no results
1	usage, deck not found, deck name too long, cannot write <print file>	bad arguments or a missing file
2	nastran: <stem> did not reach END OF JOB - see <print>	a crash; the print stops where the solver was
2	nastran: stopped after N minutes of wall clock, the limit for this run	the watchdog; the scratch directory is removed
3	nastran: the solver stopped on this message: + the first fatal line + WHAT / FIX, then nastran: <stem> finished with a FATAL message - see <print>	a USER or SYSTEM FATAL MESSAGE (either spelling; runs of blanks are collapsed before matching, so *** USER FATAL MESSAGE 3056 counts). A number with no entry prints (no explanation on file for message n; the NASTRAN User's Manual, section 6, has the list)
3	the deck was not translated, so nothing was solved (UFM 9002)	a translation fatal (UFM 9xxx names the card, file and line); no print file; read <stem>_xlat.txt
3	finished with a FATAL message with only I/O SUBSYSTEM ERROR NUMBER nnn, I/O ERROR # n ON FILE ... NAME=xxx (a buffer dump and the database directory follow) or a bare ERRTRC CALLED in the print	GINO could not read or write a block, or a module stopped without saying why; such runs used to exit 0 with END OF JOB and no results. Keep the deck and <stem>.log

exit	console line	meaning
3	nastran: this deck is written in MSC Nastran's dialect (SOL 1xx, INCLUDE, free-field cards)	you ran <code>nastran95</code> on an MSC deck
3	nastran: the solver stopped without a numbered message, which is how it runs out of open core + the -8 entry	an out-of-core stop (ERRTRC after a data block table)

*** USER POTENTIALLY FATAL MESSAGE is a note eight of NASA's demonstration decks print and run past; it is not a fatal to either scanner. Exit 0 is not "120 modes": read the `fewer modes` line (skill `nastran-analysis` §5, §8).

11. The print file

`msc/mscf06.c` rewrites the print file in place on the way out, changing layout and never a number: the eigenvector banner carries `CYCLES =` and the mode number where MSC puts them; exact zeros are `0.000000E+00`; the eigenvalue table has MSC's sub-banner ((BEFORE AUGMENTATION OF RESIDUAL VECTORS)) and no blank between header and rows; the weight generator's rows sit at MSC's columns; the sorted-echo banner is spelled as MSC spells it (the 1995 print has one more space between the words); renumbered ids are restored; modes past the number requested are cut; no line is zero-length. A line it does not recognise is copied through. Hence `VehicleDesign`'s readers (`read_nastran_eig`, `read_nastran_grids`, `read_nastran_mass`, `read_khh`, `read_nastran_flutter`, `read_nastran_pk_eigenvectors`, `read_nastran_complex_eigenvectors`, `read_nastran_aero_loads`, `read_nastran_qhhl`) read either solver's print.

A flutter print holds, per subcase: the sorted bulk data echo (`ECHO = SORT`); UIM 9457 with the sorted (Mach, k) list; the `REAL EIGENVALUES` table (a restart's spliced in from the modes run); the PK modal vectors of marked loops (and of every loop with `PKVECT`); `ROOT TRACKING` lines; the `PK METHOD FIALED TO CONVERGE` warnings; the `FLUTTER SUMMARY` blocks - with (`MATCHED POINTS`) one block per root (`POINT = n METHOD = PK`), a row per matched point with columns `KFREQ`, `1./KFREQ`, `DENSITY`, `MACH NO.`, `VELOCITY`, `DAMPING`, `FREQUENCY`, `COMPLEX EIGENVALUE` (format `F9.4` then eight `E13.6`, `mis/fa2.f` formats `1043` and `1070`); COSMIC's own PK layout is one block per (Mach, density) group with the Mach and density ratio in the header - then the `COMPLEX EIGENVECTOR` tables of the marked loops and the `AERODYNAMIC LOADS (UNIT DYNAMIC PRESSURE)` tables when `AEROF` is asked for. `read_nastran_flutter` reads both summary layouts.

12. The message catalogue (`msc/mscdiag.c`)

The first fatal in the print is repeated on the terminal with `WHAT` and `FIX` from a catalogue of 41 entries, each audited against `um/MSSG.TXT`, the emitting routine and, where a deck could provoke it, a run (`HALO.md`, "The fatal messages"). The numbers are NASA's, not MSC's. Condensed:

number	what	fix
300	a number or word on a card could not be read (field underlined in the print; <code>ERROR IN XRCARD ROUTINE</code> for control cards)	fix the field: two decimal points, a dangling exponent, an embedded blank; delete blank lines before <code>CEND</code>
307	a bulk card name the solver does not know (printed after the message)	NASTRAN-95 has <code>CQUAD4</code> / <code>CTRIA3</code> but no <code>CBEAM</code> , <code>CBUSH</code> , <code>RBAR</code> , <code>RBE2</code> , <code>EIGRL</code> , <code>PBARL</code> ; <code>nastran95ase</code> translates <code>CBUSH</code> , <code>RBAR</code> , <code>RBE2</code> , <code>EIGRL</code> , <code>PBARL</code> (the entry's own text claims <code>CBEAM</code> too; see §5.4)

number	what	fix
311	two cards of one type share an id	grid, element, property and material ids unique per type; load and constraint sets may share
313	a card's word count is not one its layout allows	usually an MSC trailing field (OFFT in field 9 of a CBAR) or a missing continuation
315	wrong kind of value in a field (integer for a real, MSC's SEID on a GRID)	decimal point on every real, none on integers, clear fields the 1970s layout leaves blank
316	illegal data (negative id, component digit above 6); a consequence when 300 came first	compare with the User's Manual layout
340	DISP rigid formats 10-12 need PARAM LMODES or LFREQ + HFREQ	add one of them
505	unknown executive control card, or a missing CEND	NASTRAN BANDIT=-1 must be line 1; MSC's ASSIGN s are not executive control here
507	an executive card in the wrong form (IMHERE 530: ID needs two names; 110: TIME an integer)	fix the echoed card
615	SET named by a non-integer	SET 10 = 1,2,3
617	a case control card with no value or a non-integer one; a PARAM in case control	move PARAM s below BEGIN BULK
2007	an element names an undefined grid	usually an INCLUDE left out of a sub-configuration deck
2010	an element names a missing property	add it, or include the common file
2050	a SUPORT names a grid that does not exist	fix the SUPORT1 in the MSC deck
2053	SPC = n selects a set no card carries	match the id; a free-free SUPORT run needs no SPC
2101A	a degree of freedom in two dependent sets	make the grid carrying the SPC or SUPORT the independent end of the rigid element
2140B	a grid id above 2,147,483 with SEQGP / SEQEP levels	renumber, or drop the sequence cards
2192	a rigid element names an undefined grid (the id is the RBAR's)	include the component file the joint attaches to
2200	the rigid-body mass matrix on the SUPORT freedoms is singular	rotary inertia on the CONM2 s, or the missing lumped-mass include
2215	a property names a missing MAT1	add it; sub-configurations must include the common file
2423	a degree of freedom made dependent twice	one rigid element per dependent grid; DIAG 21 maps SIL to grid
3005	a zero pivot in the named routine (KLL / MAA in FACTOR; SCRATCH2 in MCE1B: dependent MPC equations; (NONE) in SMA3A: a singular GENEL)	constrain or connect the columns UFM 3097 lists; give them mass; rewrite the MPCs
3008	a module ran out of open core (ADDITIONAL CORE REQUIRED when known)	raise OCMEM; rebuild with a larger NASTRAN_OPEN_CORE_WORDS
3019	MAXLINES exceeded	MAXLINES = 999999 (nastran95ase does), or print less

number	what	fix
3031	a selected set is not in the bulk data (SLT the static LOAD, DLT the DLOAD, FRL , TRL , EED , NLFT)	a set holding only SPCD is never found: add a zero FORCE
3032	METHOD = n found no EIGR of that id (EED), or a DEFORM set	match the ids; nastran95ase writes METHOD from the EIGRL
3037	the generic JOB TERMINATED IN SUBROUTINE xxxx ; from SCHEME it is BANDIT	read the message above it first; NASTRAN BANDIT=-1 or BANDTDIM=n
3056	a GRAV load with no mass matrix	RHO on the MAT1 , CONM2 s or NSM ; or take the GRAV out
3097	the named block is singular in the listed columns (KLL : no stiffness or a mechanism; MAA : GIV/MGIV on massless rotations)	SPC1 or connect; PARAM AUTOSPC 1 ; on MAA use FEER, OMIT1 456 , or add I11/I22/I33
6206	the SOF's PASSWORD is not the one it was created with	use the earlier password; one SOF per phase set
3118	a CROD / CONROD of zero length	a spring between coincident points is a CELAS2
3145	a constraint card with a blank or zero component	put the component digits in
3147	two SPC cards on one component, one with an enforced displacement	one SPC per component; plain fixities on SPC1
3176	a CBAR cannot form its axes (coincident ends, or the orientation vector along the bar)	a vertical spar gets 0. 1. 0.
3178	a LOAD combination names a set no load card carries	fold LOAD s into one; select SPCD sets directly
3179	a LOAD lists one set twice	fold the factors
2386	FEER's shifted stiffness still singular after two shifts of 100x: freedoms with neither stiffness nor mass	constrain, give mass, or ASET / OMIT them
2391	FEER's reduction produced no usable rows; follows a start vector with no positive mass norm	look for UWM 2394; rotary inertia; no negative CONM2 mass
2395	FEER's QR got a NaN or ran 200 sweeps per eigenvalue plus 1,000 (NASA's loop had no bound)	as 2391; without a 2394, report the deck (DIAG 16)
GINO	I/O SUBSYSTEM ERROR NUMBER nnn : a file read as written wrong, disk or file slots exhausted	a solver logic defect unless the scratch disk is full; report with the deck and .log
-8	out of open core with no number printed	as 3008

12.1 The front end's own messages (9000 series, msc/mscmsg.c)

Three-part shape (what, where, fix); ranges: 9000-9099 reading the deck, 9100-9199 executive and case control, 9200-9299 bulk data, 9300-9399 the model as a whole, 9400-9499 SOL 200 and the SOL 145 driver, 9500-9599 results. The ones a run meets:

number	message
UIM 9000	reading <deck> in the MSC dialect
UIM 9001	the translation summary (SOL n became rigid format r ... c cards written ...); also a continuation with no card before it (UWM)
UFM 9002	the deck was not translated, so nothing was solved
UFM 9010-9013	an INCLUDE whose name cannot be read; nested too deep; a file that cannot be opened; no bulk data (BEGIN BULK missing)
UIM 9101 / UWM 9102 / UIM 9103-9104 / UWM 9105	case control dropped (MSC's, with its cost) / unknown / options dropped, APRES to AEROF / a THRU range bounded by a renumbered id
UFM 9110 / 9111 / 9112	a SOL with no 1970s equivalent / no recognised SOL / cannot write the translated deck
UIM 9113 / 9114 / UWM 9115 / UIM 9116 / 9117 / 9119	BANDIT=-1 written / DISPLACEMENT = ALL added / several subcases, the first run / EIGC to HESS / IMETH to L / GUSTAERO sign flipped
UWM 9120, UIM 9121, UFM 9122, UWM 9123, UIM 9124, UWM 9125-9129, UIM 9130-9131, UWM 9132	OUTPUT4: an unreadable ASSIGN; another ASSIGN ignored; an unknown data block; a unit never written to; the mapping; wiring, missing or malformed files; the converted matrix; the mode cut; a short column
UIM 9123	FLUTTER: PKNL became PK with PARAM PKMATCH 1 (the number is shared with the OUTPUT4 warning above)
UWM 9133 / UIM 9133, UWM 9134	CONM2s without rotary inertia (msc/mscxl1at.c) / OUTPUT4 per-subcase files written (msc/mscop4.c) - the same number serves both / no child wrote the OUTPUT4 file
UFM 9203, UIM 9209, UIM 9211, UIM 9214, UFM 9206, UFM 9299	a non-prismatic CBEAM; PBARL to PBAR; EIGRL to EIGR FEER; SUPORT1 to SUPORT; a separated CBUSH with K2..K6; an unknown bulk card
UIM 9301, 9302	freedoms constrained for want of stiffness; ids renumbered
UIM 9400	SOL 200
UIM 9449, 9450, UFM 9451, 9452, UWM 9453, 9454, UIM 9455, UIM/UWM 9456	the SOL 145 driver: one child per subcase; the job count; a subcase not translated; a child that could not start; a child without a print; the joined print not rewritten; the child's Mach; the loads velocity
UIM 9457	from APD in the solver's print: the sorted (Mach, k) list
UFM 9460, UIM 9461-9464, UFM 9465, UWM 9466, 9467	restart: deck, dictionary or tape missing; the restart described; the dictionary lines; the cards dropped and kept; scr=no checkpointing; the tape could not be placed; the modes print missing; non-aero cards that will re-solve the modes
UIM 9470	NASTRAN SYSTEM(270) : which doublet-lattice kernel runs

Messages wrap at 72 columns on a space, a word wider than the margin (a path) left whole.

13. Troubleshooting

- Exit code and the console first.** The first fatal is repeated with WHAT/FIX. UFM 307 is a card name the 1970s solver does not know; a fatal on nastran95 followed by "this deck is written in MSC Nastran's dialect" means you wanted nastran95ase; UFM 9299 a card the front end does not know (CBEAM is one: use CBAR); UFM 9206 a separated CBUSH with K2..K6; SFM 3008 out of open core, raise OCMEM; UFM 2007 / 2192 / 2010 / 2215 / 2053 an INCLUDE left out of a sub-configuration deck.
- <stem>_xl1at.txt.** "dropped: MSC only" is fine; "dropped: no stiffness" should be only the vis panels; a large count of "degrees of freedom constrained for want of stiffness" means a component is floating (a

missing joint include, a wrong SPC id); "ids renumbered" must include `PARAM,GRDPNT` when the deck has it.

3. `<stem>_n95.dat` is the deck actually solved. When a number disagrees with Simcenter, diff it against VehicleDesign's reference translator `msc_deck_to_nastran95(deck)` card by card; a value-level diff found the eight-column truncation bug in minutes.
4. **ERRTRC CALLED with no numbered message**: out of open core; set `OCMEM` higher.
5. **Fewer modes than asked (UWM 2390)**: the mass matrix is semi-definite (UWM 2394, 9133). The modes found are valid. Ask for fewer, or give the lumped masses rotary inertia - and say why rather than doing it silently. `DIAG 16` prints the FEER reduction row by row.
6. **A flutter run with no summaries and exit 0** is a thing of the past: GINO's `I/O ERROR #` and `ERRTRC` now count as fatal. If a child dies at a late loop with `BLOCK NUMBERS INCONSISTANT ON OPEN IN DBMMGR`, `DBMEM` is set in the environment: unset it.
7. **AEROF tables full of 1e-19, or gust loads that do not change with wg**: an executable older than fork a262158 (NASA's `UNPACK` limit at row 65,536). Rerun with the current build; any model past about 10,900 grids is suspect on an older one.
8. **Jagged PSD response above 1.5 Hz**: an executable older than fork 09c8681 (the k spline in single precision).
9. **A turbulence deck stopping with UFM 316 on its TABLED1**: older than fork d5e0e5f (cards truncated at 256 fields).
10. **The export files are empty after a restart run**: the `OUTPUT4` alter is not executed on a modified restart; run the flutter deck cold (section 9).
11. **A crossing that moves between two builds or flows**: look for a one-point spike on the V-g plot first (section 7.6), then compare the (Mach, k) list (UIM 9457) and the kernel (UIM 9470): the quartic and parabolic kernels differ by 0.06-0.35 % in the monarch's crossings and the OpenBLAS LU moves marginal roots. `N95_AJJ_SOLVE=builtin` with `N95_DLM_QUARTIC=0` reproduces Jon's `halo-ase-sol145` print line for line.
12. **GINO I/O SUBSYSTEM ERROR NUMBER 102 ... AJJL on runs side by side**: the scratch disk (or tmpfs) is full. Budget runs x 5 children x 2-4 GB plus the cache; set `TMPDIR`; delete `n95_<pid>` folders of dead processes (check the process is gone).
13. **A watchdog stop on a big deck**: `N95_TIMEOUT=<minutes>`; the tests set 3 minutes so a regression fails instead of hanging. On Linux limits past 71.6 minutes wrapped until fork 9ba11cc (`usleep`'s 32-bit argument); `nanosleep` now.
14. **nastran resolves to the wrong solver**. VehicleDesign's `nastran.bat` shim runs `nastran95ase.exe` when its folder is first on the `PATH`, which shadows Simcenter's `nastran` for that shell; undo it when done. Never create a `nastran95.bat` next to the exe (cmd resolves a bare name to the `.exe` first).
15. **Profiling**: not from `<stem>.log`'s CPU column. Use `GFORTRAN_UNBUFFERED_ALL=y` and time the `BEGN/END` lines, or skill `nastran95-performance`'s `scripts/`.

14. Worked examples

From VehicleDesign's repo root on Windows (`cmd`); on Linux replace `.exe` with the bare name and `\` with `/`:

```
cd ase\vibe
..\..\utilities\open_source_software\NASTRAN\nastran95ase.exe monarch_ff_asm8083_vibe.dat
    outputs beside the deck (what ZAERO wants: <config>_vibe.out, .phg, .mzg)
..\..\utilities\open_source_software\NASTRAN\nastran95ase.exe monarch_gvt_gvt.dat C:\runs
    outputs in C:\runs; the repo stays clean
..\..\utilities\open_source_software\NASTRAN\nastran95ase.exe mfp3p2_vibe.dat out=C:\runs scr=yes
    MSC launcher spellings
```

A flutter study with the cache and the Mach children one at a time, and Jon's print reproduced (skill nastran95-performance §3):

```
cd ase\flutter\nastran_monarch_demo_asm1083
set EXE=..\..\..\utilities\open_source_software\NASTRAN\nastran95ase.exe
%EXE% --version
set N95_AERO_CACHE=D:\n95cache
set N95_TIMEOUT=240
%EXE% monarch_demo_asm1083_flutter.dat C:\temp\run1
rem the same deck as Jon's halo-ase-sol145 build prints it
set N95_AJJ_SOLVE=builtin
set N95_DLM_QUARTIC=0
%EXE% monarch_demo_asm1083_flutter.dat C:\temp\as_jon
rem every speed-up off, one Mach at a time
set N95_JOBS=1
set OMP_NUM_THREADS=1
set N95_PK_THREADS=1
set N95_DLM_THREADS=1
set N95_AMG_BATCH=0
set N95_AMP_PIPE=0
set N95_BLAS_THREADS=1
set N95_ISA=0
%EXE% monarch_demo_asm1083_flutter.dat C:\temp\serial
```

The modes once, then the flutter deck off them (the `run_nastran95ase.bat` / `run_nastran95ase.sh` in every flutter run folder is these two lines):

```
nastran95ase monarch_demo_asm1083_modes.dat scr=no
nastran95ase monarch_demo_asm1083_flutter.dat restart=monarch_demo_asm1083_modes.dat
```

From MATLAB (the shape the tests use; `call` keeps `cmd.exe` from eating the quotes):

```
set_up_path
exe = fullfile(VehicleDesign_root, 'utilities', 'open_source_software', 'NASTRAN',
'nastran95ase.exe');
deck = fullfile(VehicleDesign_root, 'ase', 'vibe', 'monarch_gvt_gvt.dat');
outd = tempname; mkdir(outd);
[rc, cmdout] = system(sprintf('call "%s" "%s" "%s"', exe, deck, outd));
assert(rc == 0, cmdout); % 1 bad args, 2 no END OF JOB, 3 FATAL
prt = fullfile(outd, 'monarch_gvt_gvt.out');
[f, phi, nodes] = read_nastran_eig(prt);
```

A NASA demonstration deck through the 1970s executable:

```
nastran95 inp\d10021a.inp C:\runs\demos
```

The fast check of the whole SOL 145 path is VehicleDesign's `test_nastran95ase_flutter_subcases.m` (a cantilever with one CAERO1 panel, two Machs, three matched points each, the third marked, through the driver in a second, and the same deck restarted off its modes checkpoint); the monarch deck's lowest crossing per Mach is held by `test_nastran95ase_aeroelastic.m` (skill `build-nastran95`).